Correlation Between Lymphocytes and Autophagy

cf. IF v3.7.Rmd

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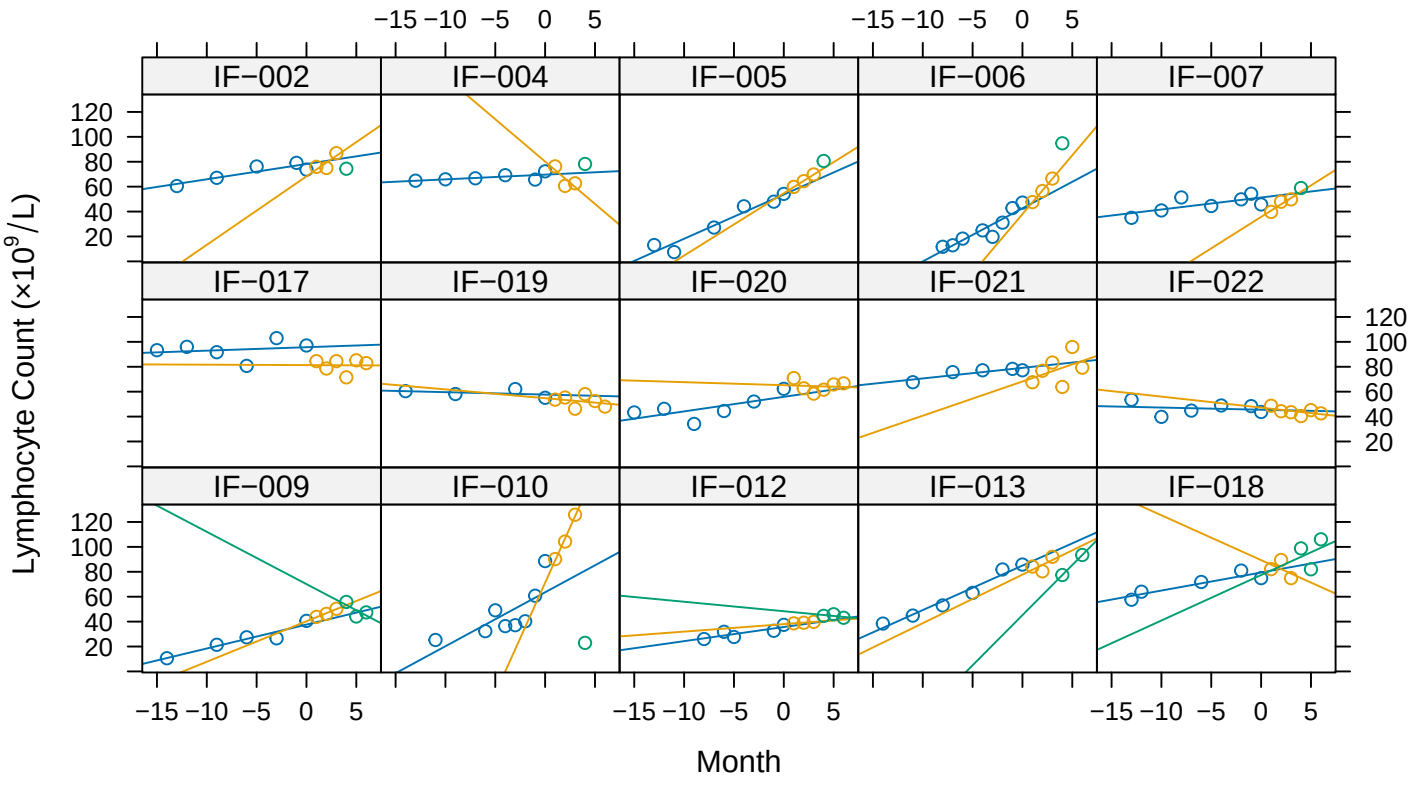
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1. Data description (same as IF v3.7.Rmd p. 1)

The longitudinal data are from an experiment to study the effect of intermittent fasting (IF) on the lymphocyte measurements of 15 participants, each of whom was repeatedly measured before, during, and after the intervention. The data set (long format) consists of six columns:

- **LYMP** - [numeric] lymphocyte count ($\times 10^9$/L); lower values usually correspond to better outcomes;
- **ATG** - [numeric] relative net LC3-II expression; higher levels are associated with increased autophagy;
- **ID** - [factor] participant identifier; 15 levels;
- **Time** - [numeric] centered time point, with a frequency of 30 days, treating the **last** pretest as Month 0;
- **Time_Within_Phase** - [numeric] time point within each phase, starting from Month 0;
- **Phase** - [numeric] treatment identifier; 0 as pretest, 1 as intervention, and 2 as posttest.

```
df <- read.csv("~/Downloads/IF_v3.1.csv", stringsAsFactors=T)
lattice::xyplot(LYMP ~ Time | ID, data=df, group=Phase, type=c("p","r"), layout=c(5,3),
                index.cond=list(c(6:9,11,10,12:15,1:5)), #change panel order
                xlab="Month", ylab=expression(paste("Lymphocyte Count (\u00d7",10^9/L,")")))
```



2. Modeling

The additional study aims to examine the correlation between lymphocyte count and relative net LC3-II expression, using a linear mixed-effects model once again.

Given the small sample, we would report the raw-scale model if its relationship appears reasonably linear and is not dominated by the highest MLC observations. It is simpler and clinically easier to interpret. The log model is preferable if the association clearly flattens at higher counts or the largest counts have excessive leverage.

```
dat <- df[!is.na(df$LYMP) & !is.na(df$ATG) & df$Phase==1,] #omit missing values
dat <- droplevels(dat)

fast6 <- paste0("IF-0",c(17,19,20,21,22)) #cohort definition used in IF-v3.7.Rmd
dat$Fast <- factor(ifelse(dat$ID %in% fast6, "Six", "Three"))

fit.ATG_LYMP <- nlme::lme(
  fixed  = ATG ~ LYMP + Time_Within_Phase + Fast,
  random = ~ 1 | ID, #random intercept for each subject
  data   = dat,
  method = "REML"
)

summary(fit.ATG_LYMP)
```

```
## Linear mixed-effects model fit by REML
##   Data: dat
##       AIC      BIC    logLik
##   25.7675 36.60748 -6.883752
##
## Random effects:
##  Formula: ~1 | ID
##         (Intercept)  Residual
## StdDev:   0.2722734 0.1729085
##
## Fixed effects:  ATG ~ LYMP + Time_Within_Phase + Fast
##                        Value  Std.Error DF   t-value p-value
## (Intercept)        0.9949508 0.22851395 34  4.354005  0.0001
## LYMP              -0.0055189 0.00297756 34 -1.853497  0.0725
## Time_Within_Phase -0.0494235 0.01972953 34 -2.505054  0.0172
## FastThree         -0.0505441 0.16897708 11 -0.299118  0.7704
##  Correlation:
##                   (Intr) LYMP   Tm_W_P
## LYMP              -0.804
## Time_Within_Phase -0.092 -0.151
## FastThree         -0.296 -0.195  0.205
##
## Standardized Within-Group Residuals:
##        Min         Q1        Med         Q3        Max
## -1.3818744 -0.6312129 -0.1842153  0.5467234  2.3823940
##
## Number of Observations: 49
## Number of Groups: 13
```

```r
coef(fit.ATG_LYMP) #the coefficients across 13 subjects
```

```
##        (Intercept)         LYMP Time_Within_Phase   FastThree
## IF-002   0.7987660 -0.005518897       -0.04942353 -0.05054407
## IF-004   0.7236320 -0.005518897       -0.04942353 -0.05054407
## IF-005   0.8908506 -0.005518897       -0.04942353 -0.05054407
## IF-006   1.2201955 -0.005518897       -0.04942353 -0.05054407
## IF-009   0.6304075 -0.005518897       -0.04942353 -0.05054407
## IF-010   1.5364123 -0.005518897       -0.04942353 -0.05054407
## IF-013   1.2039246 -0.005518897       -0.04942353 -0.05054407
## IF-017   0.9170463 -0.005518897       -0.04942353 -0.05054407
## IF-018   0.9554181 -0.005518897       -0.04942353 -0.05054407
## IF-019   0.8488253 -0.005518897       -0.04942353 -0.05054407
## IF-020   0.9719149 -0.005518897       -0.04942353 -0.05054407
## IF-021   1.1392596 -0.005518897       -0.04942353 -0.05054407
## IF-022   1.0977082 -0.005518897       -0.04942353 -0.05054407
```

```r
nlme::intervals(fit.ATG_LYMP) #the 95% confidence interval for the fixed effect
```

```
## Approximate 95% confidence intervals
##
##  Fixed effects:
##                           lower         est.        upper
## (Intercept)         0.53055463  0.994950843  1.459347054
## LYMP               -0.01157002 -0.005518897  0.000532230
## Time_Within_Phase -0.08951876 -0.049423534 -0.009328304
## FastThree          -0.42246011 -0.050544070  0.321371974
##
##  Random Effects:
##   Level: ID
##                       lower      est.     upper
## sd((Intercept)) 0.1558346 0.2722734 0.4757145
##
##  Within-group standard error:
##     lower      est.     upper
## 0.1348156 0.1729085 0.2217648
```

To examine the association between absolute lymphocyte count and autophagy during the time restricted eating (TRE) intervention, we fitted a linear mixed-effects model with relative net LC3-II expression as the outcome and a participant-specific random intercept. The analysis included 49 observations from 13 participants.

After accounting for time within the intervention phase and TRE cohort, higher lymphocyte counts were associated with lower relative net LC3-II expression, although the confidence interval included zero ($\beta_1 = -0.0055$ per 10^9/L, 95% CI $[-0.0116, 0.0005]$, $t_{34} = -1.85$, $p = 0.0725$). Thus, the adjusted analysis suggested a negative association between lymphocyte count and autophagy, but the data did not provide sufficiently precise evidence to conclude that this association differed from zero.

Relative net LC3-II expression decreased significantly over time during the intervention ($\beta_2 = -0.0494$ per month, 95% CI $[-0.0895, -0.0093]$, $t_{34} = -2.51$, $p = 0.0172$).

The estimated difference between the 3- and 6-month cohorts was small and imprecise ($\beta_3 = -0.0505$, 95% CI $[-0.4225, 0.3214]$, $t_{11} = -0.30$, $p = 0.7704$).